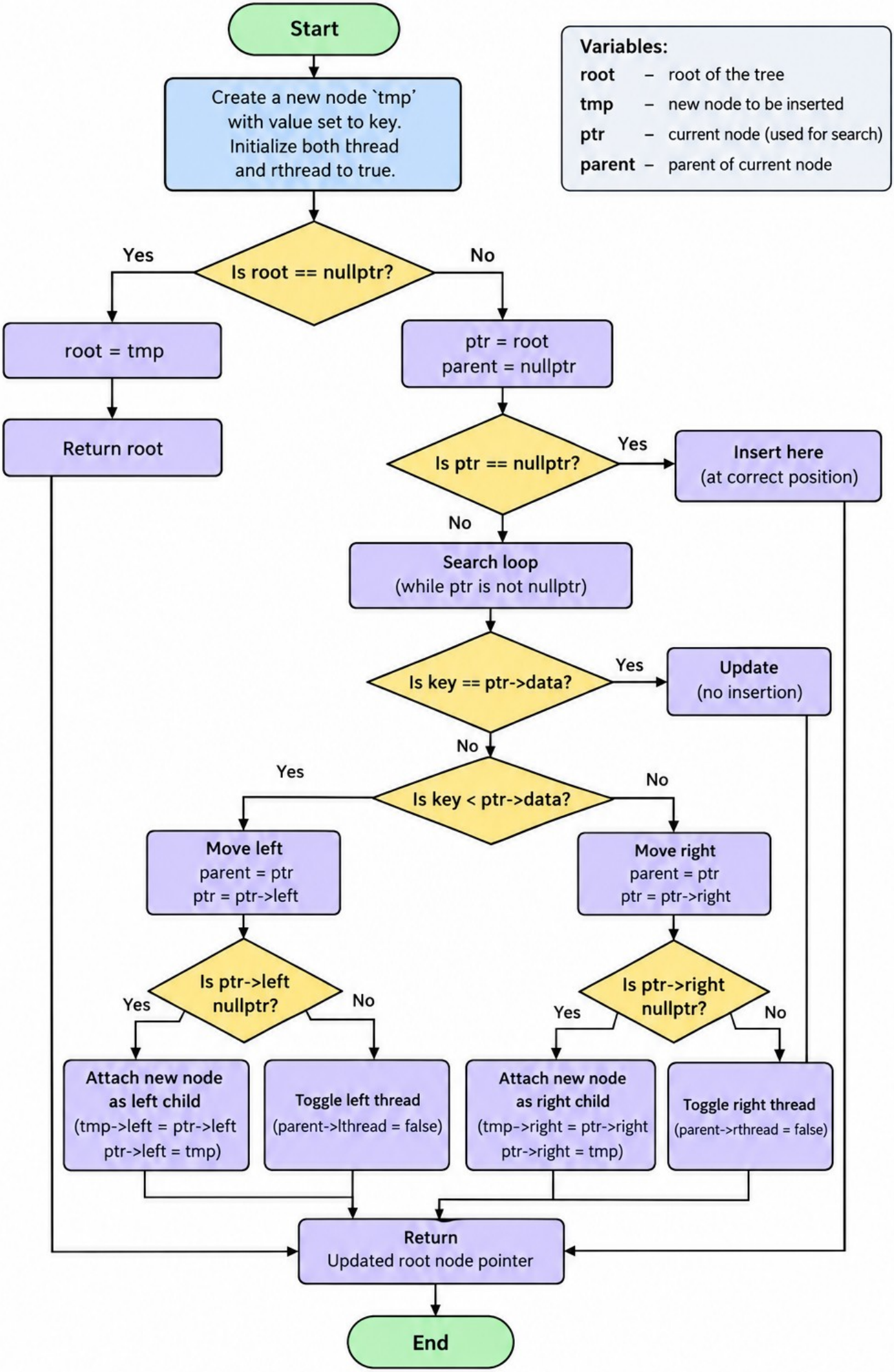

Start
Create a new node `tmp' with value set to key. Initialize both lthread and rthread to true.
Variables:
root – root of the tree
tmp – new node to be inserted
ptr – current node (used for search)
parent – parent of current node
Is root == nullptr?
Yes
No
root = tmp
Return root
ptr = root
parent = nullptr
Is ptr == nullptr?
Yes
Insert here (at correct position)
No
Search loop (while ptr is not nullptr)
Is key == ptr->data?
Yes
Update (no insertion)
No
Is key < ptr->data?
Yes
No
Move left
parent = ptr
ptr = ptr->left
Move right
parent = ptr
ptr = ptr->right
Is ptr->left nullptr?
Yes
No
Is ptr->right nullptr?
Yes
No
Attach new node as left child
(tmp->left = ptr->left
ptr->left = tmp)
Toggle left thread
(parent->lthread = false)
Attach new node as right child
(tmp->right = ptr->right
ptr->right = tmp)
Toggle right thread
(parent->rthread = false)
Return
Updated root node pointer
End